DATA 202 — Ensembles · Week 11 Wednesday

Same pumpkin seeds as Monday. Ç = Çerçevelik, Ü = Ürgüp Sivrisi. Your professor will tell you when to do each part.

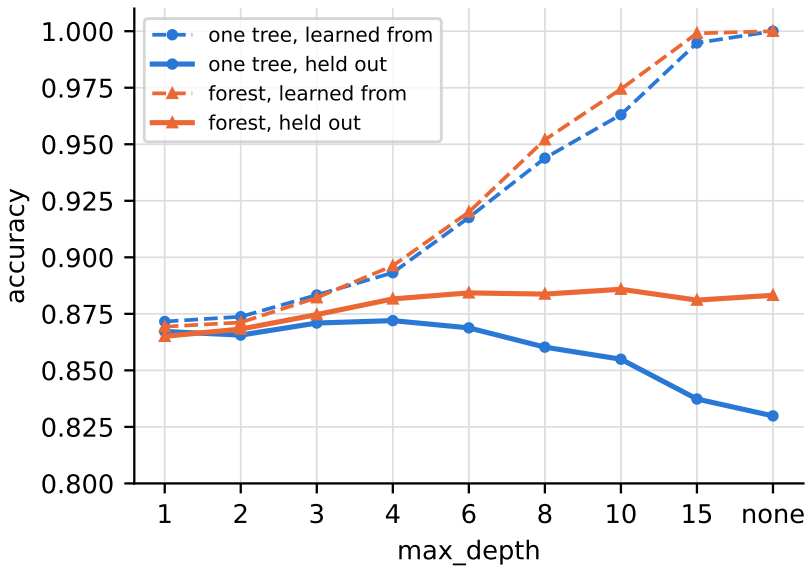
1 · Five trees vote

A random forest of five deep trees, each grown on its own bootstrap sample of the training seeds. Their votes on six new seeds:

seed	tree 1	tree 2	tree 3	tree 4	tree 5	forest	actual	right?
A	Ü	Ü	Ü	Ü	Ü	_____	Ü	_____
B	Ü	Ç	Ç	Ç	Ç	_____	Ç	_____
C	Ç	Ç	Ç	Ü	Ü	_____	Ç	_____
D	Ç	Ç	Ü	Ç	Ç	_____	Ç	_____
E	Ü	Ç	Ü	Ü	Ç	_____	Ü	_____
F	Ü	Ü	Ü	Ü	Ü	_____	Ç	_____

1. **Fill in** the forest's answer (the majority vote), then mark it right or wrong.
2. **Circle** every wrong vote of a single tree. Is there a tree with no circles? _____
3. **Pair talk:** which single tree would you trust alone? Would 100 more trees fix seed F?

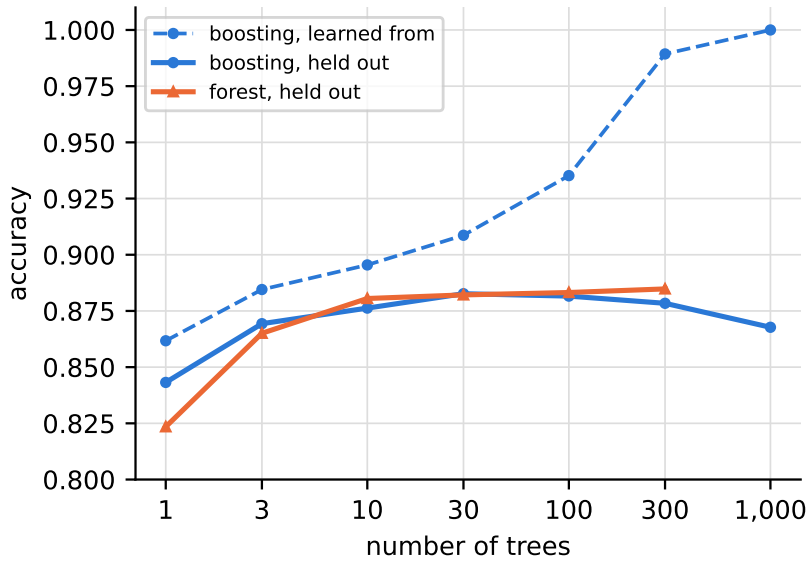
2 · Tree vs. forest



Five-fold cross-validation on the training seeds, as the trees grow deeper.

1. **Label** the side where the single tree overfits.
2. **Pair talk:** both reach 100% on the seeds they learned from. Why doesn't the forest's held-out accuracy fall?

3 · More trees



Five-fold cross-validation on the training seeds, as we add trees. Boosting: small trees in a row, each learning from the mistakes left by the ones before it.

1. **Circle** where boosting starts to overfit.
2. **Pair talk:** why can more trees hurt boosting but not the forest?